

Robust Generalization, Rigorous Evaluation, and Impactful Applications of Large Language Models

Large language models (LLMs) have achieved remarkable, and in some domains superhuman, capabilities by scaling data, parameters, and compute. Yet they remain brittle under distribution shift, revealing a core limitation of scaling: strong in-distribution performance does not guarantee robust real-world generalization. My research tackles this gap by treating LLM reasoning as inference from noisy observations to unobserved structures (*e.g.*, abstractions, causes, and latent concepts). I advance this agenda through three tightly connected directions: studying **robust generalization** under extreme linguistic distribution shifts (§1), developing **rigorous evaluation** frameworks that account for distributional, cultural, and uncertainty-based confounds (§2), and grounding LLMs in **impactful, expert-centric domains** where uncontrolled, in-the-wild shifts are unavoidable (§3). Ultimately, I aim to move beyond scale-driven learning toward robust, reliable, and inclusive LLMs that support scientific discovery and high-stakes decision-making.

1. Robust Generalization under Extreme Distribution Shift

The first pillar of my research focuses on how LLMs generalize beyond their training distributions. I use code-switching, a common linguistic phenomenon where multiple languages are interleaved within a single context, as a controlled yet extreme setting to probe generalization, inducing distribution shifts along lexical, syntactic, and safety dimensions simultaneously. I aim to (a) **empirically characterize the emergent capabilities and systematic failures of LLMs** under such shifts [1–3] and (b) **design human-inspired inductive biases of LLMs that enable more robust generalization** [4, 5].

Superhuman yet Fragile Generalization. I show that LLMs exhibit **multilingual generalization that surpass typical human exposure**. While humans generally use one or two languages, LLMs can understand, reason about, and generate code-switching between 15+ language pairs [1]. Furthermore, they can process complex inter- and intra-sentential code-switching among 10+ languages simultaneously [2]. These findings demonstrate that large-scale multilingual training induces emergent abilities beyond human-scale exposure. At the same time, I **identify systematic and unhumanlike failure modes of LLMs**. Through code-switching red-teaming (CSRT) [2], I reveal that safety alignment is highly inconsistent across languages: models that behave safely in English often produce undesirable outputs in low-resource languages due to skewed alignment data. In OLA [3], I further show that LLMs struggle to respond in the expected language given code-switching inputs, highlighting brittleness to input patterns that are rare in training [3]. Together, these results expose a sharp gap between superhuman capability and robust generalization.

Human-Inspired Inductive Biases for Robust Generalization. To address these limitations, I **propose incorporating human-inspired inductive biases that explicitly guide how models should generalize beyond observed distributions**. Rather than relying solely on scale-driven emergence, I demonstrate how structured inductive bias can improve robustness with greater efficiency. In CSCL [4], I introduce a data-efficient continual pre-training method that uses code-switching corpora as a structured curriculum, inspired by human second-language acquisition. This structured curriculum stabilizes multilingual representations and improves robustness under linguistic distribution shifts. In CSICL [5], I extend this principle to inference-time, showing that progressive code-switching as in-context learning enables effective adaptation without additional training. These results demonstrate that carefully-designed inductive biases can improve generalization and reduce failure modes, while preserving the superhuman capabilities of LLMs.

2. Rigorous Evaluation under Distribution, Language, and Uncertainty Shift

As LLMs are deployed across increasingly diverse settings, evaluation itself has become a major bottleneck. A second pillar of my research focuses on **identifying and correcting systematic distortions in evaluation**

practices that arise from implicit assumptions about data distributions [6, 7], languages/cultures [8–11], and uncertainty [12, 13]. These evaluation practices have received wide attention (*150+ citations in total*), *adopted by state-of-the-art multilingual and Korean-specialized LLMs* (e.g., Phi-3.5 from Microsoft, Command R+ from Cohere, HyperCLOVA X (THINK) from NAVER, and K-EXAONE from LG).

Distribution-Aware Evaluation. I show that many widely used LLM benchmarks are heavily skewed toward specific languages, domains, and task formulations, often distorting conclusions about model capability. To address this, I introduce **BenchHub** [6], a unified, customizable, and dynamic benchmark suite, supporting ten languages. I extend it to **BenchHub+** [7], an interactive leaderboard that reveals how model rankings change under different evaluation configurations. This work reframes evaluation as a conditional process rather than a fixed score.

Language- and Culture-Aware Evaluation. Beyond distribution coverage, I examine how cultural and linguistic assumptions embedded in benchmarks affect evaluation validity. I demonstrate that naïve translation of English benchmarks fails to capture culturally grounded bias. In **KoBBQ** [8], I propose a general pipeline for culturally adapting benchmarks, recognized as *Oral Presentation at ACL* and *extended to eight subsequent works* adapting KoBBQ into 15 languages. Complementing this, I release **CLiCK** [9] to evaluate Korean linguistic and cultural competence (*34k+ data downloads*). More recently, I construct **KCaQA** [10], open-ended question-answer pairs covering Korean cultures in four languages. These efforts directly inform my contributions to the development of **HyperCLOVA X** [11], a Korean-specialized LLM.

Uncertainty-Aware Evaluation. Finally, I investigate how ambiguity in benchmark design conflates *data uncertainty* with *model uncertainty*. To disentangle them, I introduce **MAQA** [12], a dataset explicitly constructed to remove ambiguity, enabling principled evaluation of uncertainty, calibration, and confidence. I further analyzed how uncertainty manifests in layer-wise inference dynamics, showing that uncertainty does not substantially alter where models commit to predictions during decoding [13].

3. Impactful Applications of AI for Unlocking Expert-Only Domains

Beyond controlled benchmarks, **I study how LLMs can be harnessed to advance expert-only domains** by lowering access barriers, scaling expert reasoning, and enabling new paradigms. **These domains constitute uncontrolled, in-the-wild distribution shift** with non-standard language, domain-specific assumptions that rarely appear in observed data, and long-tail edge cases. Grounding LLMs in such settings maximizes real-world impact, while stress-testing and improving robustness where generalization matters most.

AI for Historical Discovery. Historical discovery has long relied on a handful of experts due to the use of archaic languages and the high cost of specialized background knowledge. I systematically lower these barriers while accelerating expert workflows through language modeling. In particular, I pioneer language modeling in Hanja [14–18], an ancient Korean writing system based on Chinese characters. In **HUE** [14], I introduce the first evaluation benchmarks and pretrained language models for Hanja. In **OKHC** [15], I release the first large-scale Korean historical corpora, spanning from the seventh century to the present. Building on these resources, I identify fundamental limitations of standard NLP adaptation techniques for historical documents and propose optimized Hanja language modeling methods [16]. In **H2KE** [17], I build neural machine translation systems that translate Hanja documents into modern Korean and English. In **HERITAGE** [18], I finally integrate these models into a web platform that enables human-in-the-loop workflows for both historians and the general public. Together, these contributions establish a foundation for community-driven, large-scale historical discovery that was previously infeasible at scale.

AI for Education Innovation. I also study how generative AI can transform language education by reducing over-reliance on human instructors, while enabling **(a) interactive feedback** [19–23] and **(b) personalized learning** [24–26] (*250+ citations in total*). To support interactive learning at scale, I develop

AI-assisted platforms for English as a Foreign Language (EFL) writing [19] and speaking [20], deploying them to over 200 college students. In **RECIPE4U** [21], I publicly release multi-semester interaction logs and edits, enabling reproducible research on student–AI interactions. I demonstrate how AI can simultaneously support learner engagement, gamify practice, and generate reliable NLP annotations [22]. Complementing these student-facing systems, I design a prompt analytics dashboard that helps instructors interpret and monitor student–AI interactions [23]. For personalized learning, I focus on data-driven pedagogy. In **DREsS** [24], I release the first large-scale, rubric-based automated essay scoring (AES) dataset for EFL writing and develop LLM-based AES models (*200+ data requests*). Building on this foundation, I introduce a score-conditioned feedback generation framework that produces more actionable and pedagogically aligned feedback [25]. More recently, I propose a transcreation-based pipeline for generating personalized reading materials [26], recognized as *Best Paper Award at EMNLP workshop*.

4. Future Directions

Building on my findings that LLMs can be superhuman yet brittle under distribution shift, my future work targets generalization beyond observed data as a foundation for reliable reasoning. I will develop robust generalization across languages, modalities, and domains and ground them in scientific settings to enable reliable and inclusive decision making in high-stakes domains.

Robust Generalization across Languages. First, I will extend code-switching-based, human-inspired inductive biases into diverse training phases of LLMs. Beyond one-to-one language transfer, I will introduce *simultaneous cross-lingual transfer*, in which structurally related languages are interleaved within a unified continual learning curriculum that reflects shared morphology, syntax, and script. This approach aims to reinforce superhuman generalization of LLMs that can understand multiple languages simultaneously. To effectively handle unseen languages and tasks at inference time, I will design multilingual test-time scaling methods with verifiable rewards by adaptively switching latent reasoning languages according to context. This approach is inspired by human bilingual behavior, where speakers naturally associate specific concepts and abstractions with particular languages.

Robust Generalization beyond Languages. I will further extend these principles to broader forms of heterogeneity in the real-world, including modalities and domains. Across modalities (*e.g.*, text–speech, text–image), I will study how structured mixing of modalities during training improves generalization in data-efficient multimodal learning. Across domains, I will investigate domain-aware mixing and controlled intra-document switching as a principled data augmentation strategy. Taken together, this line of research treats heterogeneity as a *controllable axis of supervision*: by exposing models to structured and partially shifted contexts, I aim to induce representations that are invariant to superficial variation while remaining sensitive to task-relevant structure. I will develop practical training and test-time procedures that regulate the degree of mixing (*e.g.*, composition ratios and switching schedules), and systematically evaluate robustness under distribution shifts spanning modality gaps, domain drift, and their interactions.

AI for Scientific Discovery. Finally, I will extend this generalization to scientific reasoning, where abstraction, uncertainty, and incomplete evidence are intrinsic. To ground this investigation, I will curate multimodal (*i.e.*, text–speech–image) scientific resources from high-stakes, real-world domains (*e.g.*, climate, healthcare), sourced from local, underrepresented communities. This effort will center on knowledge preserved through oral traditions, proverbs, and archival documents, which remain largely absent from Western-centric LLMs. Building on these resources, I will develop methods that enable LLMs to infer unobserved scientific structure, reason from partial observations, and generate scientific yet inclusive explanations, with the goal of enabling reliable and globally representative scientific discovery.

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